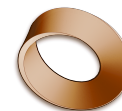


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Design



3a.6–10 – Your Roadmap to Level 10

Unit 3a: Interactive Animations & Games

Learning intentions

Five lessons, one goal: reach **Level 10** by **Friday 2 October**, while thinking about the greeting card you'll build with these same skills.

HOMEWORK

Finish code.org Unit 3a through **Level 10**, due **Friday 2 October**.

6 Random Numbers

`randomNumber()` makes your program do something different every time it runs. You'll update a variable each time round a loop and draw a picture that's a little different every time you press play.

Card thinking: randomness is a tool, not a requirement — your card can use it for a scattered background, or not at all.

7 Robot Faces (mini-project)

Your first open-ended build: design a robot face using shapes and variables. **Don't stop at rect and ellipse** — `line()`, `arc()`, `point()` and `regularPolygon()` all count too, and you need at least four different shapes.

Card thinking: this is where you practise combining several shapes into one picture — exactly what your card needs.

8 Sprites

`createSprite()` places an image on the screen. A sprite's *look* comes from a separate drawing tool, the **Animation Tab** — pencil, fill bucket, shapes — not from code.

Card thinking: only relevant if you want a drawn character on your card, rather than shapes built entirely from code.

9 Sprite Properties

Move, resize and rotate a sprite using dot notation: `sprite.x`, `sprite.y`, `sprite.rotation`, `sprite.scale`.

Card thinking: useful only if Lesson 8 gave your card a sprite to position.

10 Text

`text()` puts words on the screen; `textFont()` and `textSize()` control how they look. This is where your card's **message and title** come from — see the separate Fonts page for the full list of fonts you can use and why.

Card thinking: decide your message now. What does the card say, and in what font?

Activity

Working independently Move through these at your own pace. Stuck on one level? Try a neighbour on the same level before asking — most answers are visible on someone else's screen.